Northampton Community College
Statistical Methods
Hypothesis Testing
Homework

Follow the steps discussed in class to answer the questions below. Please add the RStudio code and output for each problem and keep the same numbering system.

1. The anterior cruciate ligament (ACL) is one of the ligaments that help stabilize the knee. Surgery is often recommended if the ACL is completely torn, and recovery time from the surgery can be lengthy. A medical center developed a new surgical procedure designed to reduce the average recovery time from the surgery. To test the effectiveness of the new procedure, a study was conducted in which 210 patients needing surgery to repair a torn ACL were randomly assigned to receive either the standard procedure or the new procedure.

    a. Based on the design of the study, would a statistically significant result allow the medical center to conclude that the new procedure causes a reduction in recovery time compared to the standard procedure for patients like those in the study? Explain your answer.

    Yes, if the results were significantly small enough then there would be grounds to conclude the new procedure would result in a reduction of recovery time. But if the results are statistically close in variation, then I believe more test would need to be done before we could make an official claim.

    b. Summary statistics on the recovery times from the surgery are shown in the table.

Type of Procedure	Sample Size	Mean Recovery Time (days)	Standard Deviation Recovery Time (days)
Standard	110	217	34
New	100	186	29

Does the data provide convincing statistical evidence that those who receive the new procedure will have less recovery time from the surgery, on average, than those who receive the standard procedure, for patients like those in the study? Use a 5% level of significance.

At an alpha level of 0.05, we concluded that there's significant evidence in the support of the "New" procedure reducing the recovery time of post ACL surgery in comparison to the "Standard" procedure.

2. The nutrition label on a brand of iced tea says that the average amount of sugar per bottle is 25 grams. A chemical analysis of a sample of 30 bottles finds an average of 33.8 grams of sugar per bottle. Test to see if this provides significant evidence that the true average is greater than 25 grams of sugar.

   At an alpha level of 0.05, we concluded that there was significant evidence in support of the true average grams of sugar per bottle of iced tea was greater than the previously found 25 grams.

3. A brain researcher introduces a study in which 40 participants are trained to solve problems in a certain way and then asked to solve an unfamiliar problem that requires fresh insight. Half of the participants were randomly assigned to receive electrical stimulation of the brain while the other half (control group) received sham stimulation (a shock to the brain with no effect) as a placebo. The results are shown in Table 4.12.

Table 4.12 ***Does electrical brain stimulation bring fresh insight to a problem?***

Treatment	Solved	Not Solved
Sham	4	16
Electrical	12	8

   Do participants who receive electrical brain stimulation solve the insight problem faster, on average, than those who receive sham stimulation? Use a 5% level of significance.

   Once the hypothesis test was ran; we found that at an alpha level of 0.05, we can conclude that there is significant evidence to support students who underwent the electrical brain stimulation had a higher likelihood of solving the problem faster than those who were apart of the placebo group. With a 95% confidence interval of (0.167 to 1)

4. A local community center recently launched a new after-school reading program designed to help middle-school students improve their literacy skills. At the end of the semester, all students took the same reading comprehension assessment. The center wants to know whether participating in the program actually increased the proportion of students who passed the assessment. Out of 80 students who attended the reading program regularly, 62 passed the assessment. Among 75 students who did not participate, 48 passed.

   Test whether the pass rate is higher for students who participated in the reading program. Use a 5% level of significance.

   After running a hypothesis test at an alpha level of 0.05, we can conclude that there is significant evidence in support of students who attended reading program regularly had a

higher passing rate in comparison to the students who did not attend. With a 95% confidence interval of (0.016 to 1).

Appedix: Problem 1

```
mu0  =  0 # hypothesized difference
x1bar =    186  # sample1 mean
 x2bar =   217   #sample2 mean
 s1 =    29 #sample1 std dev
 s2 =  34   #sample 2 std dev
 n1 =  100  #sample1
 n2 = 110   #sample2
 t = ((x1bar-x2bar)-(mu0))/sqrt((s1^2/n1)+(s2^2/n2))
```

output; -7.127079

```
# Degrees of Freedom
DF=(s1^2/n1+s2^2/n2)^2/(( s1^2/n1)^2/(n1-1)+ (s2^2/n2)^2/(n2-1))
```

Output; 207.1791

```
pval = pt(-abs(t), df=DF)
```

output; 8.357991e-12

Appendix: Problem 2

```
xbar =    33.8       # sample mean
 mu0 =  25          # hypothesized value
 s =     0.36            # sample standard deviation
```

```
  n =   30              # sample size
  t = (xbar-mu0)/(s/sqrt(n))
t                    # test statistic
```
output; 133.8877

```
pval = pt(-abs(t), df=n-1)  # using lower tail
pval                    # one-tailed test
```
output; 2.428873e-42

Appendix: Problem 3

```
alpha = 0.05
  n1 =   20   #sample1
  n2 =  20  #sample2
  succ1  =  12  #number of success for sample 1
  succ2  =  4  #number of success for sample2
  prop.test(c(succ1,succ2), c(n1,n2), conf.level=1-alpha, alternative= "greater", correct = FALSE)
```
Output; 2-sample test for equality of proportions without continuity correction

```
data:  c(succ1, succ2) out of c(n1, n2)
X-squared = 6.6667, df = 1, p-value = 0.004912
alternative hypothesis: greater
95 percent confidence interval:
 0.1673826 1.0000000
sample estimates:
prop 1 prop 2
   0.6    0.2
```

Appendix: Problem 4

```
alpha = 0.05
  n1 =   80   #sample1
  n2 =  75  #sample2
  succ1  =  62  #number of success for sample 1
  succ2  =  48  #number of success for sample2
  prop.test(c(succ1,succ2), c(n1,n2), conf.level=1-alpha, alternative= "greater", correct = FALSE)
```

output; 2-sample test for equality of proportions without continuity correction

```
data:  c(succ1, succ2) out of c(n1, n2)
X-squared = 3.4241, df = 1, p-value = 0.03213
alternative hypothesis: greater
95 percent confidence interval:
 0.01579988 1.00000000
sample estimates:
prop 1 prop 2
 0.775  0.640
```